



Module Presenter's Manual

for

Managing Large Datasets with
MongoDB

Effective from: December, 2023
Ver. 1.0

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	December 2023	New	-

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1. Introduction

At the end of this course, students will be able to:

- Master the skills of Replication and Sharding of data in MongoDB to optimize read/write performance
- Perform installation, configuration, and maintenance of MongoDB environment
- Get hands-on experience in creating and managing different types of indexes in MongoDB for query execution
- Develop skill sets in processing huge amounts of data using MongoDB tools
- Proficiently store unstructured data in MongoDB
- Gain proficiency in MongoDB configuration, backup methods as well as monitoring and operational strategies
- Acquire in-depth understanding of managing DB Notes, replica set & Master-Slave concepts

2. Information on Session Allocation

Module	Online Hours	Self-Study Hours
Managing Large Datasets with MongoDB	32	10

Throughout this Presenter's Manual, the module, **Managing Large Datasets with MongoDB**, will be referred to as **MongoDB**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables

Resources available on Onlinevarsity:

Feature - Description/Functionality
Ebook -Student reading material in the form of PDF
Download eBook - Student has the option to download the subject related e-book and read offline
Practice 4 Me - Student can test and evaluate their understanding of module related topics.
Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
References - Student can access additional subject related material for reading.

4. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 MONGODB – TL1	Session 2 MONGODB – TL2	Session 3 MONGODB –TL3	Session 4 MONGODB – TL4
2	Session 5 MONGODB –TL5	Session 6 MONGODB – TL6	Session 7 MONGODB – TL7	Session 8 MONGODB –TL8
3	Session 9 MONGODB – TL9	Session 10 MONGODB – TL10	Session 11 MONGODB –TL11	Session 12 MONGODB – TL12
4	Session 13 MONGODB – TL13	Session 14 MONGODB –TL14	Session 15 MONGODB – TL15	Session 16 MONGODB – TL16

MONGODB-Managing Large Datasets with MongoDB

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	MONGODB – TL1	All the topics as listed below from Session 1 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 1 - Introduction to MongoDB</u> - Databases - An Overview of Types of DBMS - RDBMS VS NoSQL - Overview of MongoDB - What is MongoDB and how it works - Database, Collection, and Document - Getting started with MongoDB - MongoDB Server - MongoDB Cloud - MongoDB tools - Install MongoDB Shell - Install MongoDB Compass - Connecting the server with MongoDBShell	Managing Large Datasets with MONGODB SG-Session 1 XP-Session 1 TG-Session 1
2	MONGODB – TL2	All the topics as listed below from Session 2 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 2 - Working with MongoDB Databases</u> - Data Types in MongoDB - Databases and Collections - Create Database - Create Collection - Insert Document - Query Document - Update Document - Delete Document - Drop Collection - Drop Database	Managing Large Datasets with MONGODB SG-Session 2 XP-Session 2 TG-Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
3	MONGODB – TL3	All the topics as listed below from Session 3 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 3- MongoDB Operators</u> • Introduction to MongoDB operators • Query and Projection Operator • Comparison • Logical • Element • Array • Projection • Miscellaneous • Update Operator • Field • Array	Managing Large Datasets with MONGODB SG-Session 3 XP-Session 3 TG-Session 3
4	MONGODB – TL4	The Try It Yourself questions of Session 1 to Session 3 of <i>Managing Large Datasets with MongoDB</i> book should be covered in this session.	Managing Large Datasets with MONGODB Session1 to 3
	S1	All the Assignments from Session 1 to Session 4 of <i>Managing Large Datasets with MongoDB</i> course should be covered in this session.	Managing Large Datasets with MONGODB (OnlineVarsity) Session 1-4
5	MONGODB – TL5	All the topics as listed below from Session 4 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 4 – Aggregation Pipeline</u> • Introduction to Aggregation Pipeline • Aggregation Pipeline Stages • Aggregation Expressions	Managing Large Datasets with MONGODB SG-Session 4 XP-Session 4 TG-Session 4

Session No.	Session Title	Session Details	Deliverables' Mapping
6	MONGODB – TL6	All the topics as listed below from Session 5 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 5 – Database Commands</u> • Introduction to Database Commands • Authentication commands • Write operation commands • Geospatial commands • Aggregation commands	Managing Large Datasets with MONGODB SG-Session 5 XP-Session 5 TG-Session 5
7	MONGODB – TL7	All the topics as listed below from Session 6 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 6 – MongoDB Shell Methods</u> • Introduction to MongoDB Shell methods • Collection methods • Database methods • User management methods • Role management methods	Managing Large Datasets with MONGODB SG-Session 6 XP-Session 6 TG-Session 6
8	MONGODB – TL8	The Try It Yourself questions of Session 4 to Session 6 of <i>Managing Large Datasets with MongoDB</i> book should be covered in this session.	Managing Large Datasets with MONGODB Session 4-6
	S2	All the Assignments from Session 5 and Session 6 of <i>Managing Large Datasets with MongoDB</i> course should be covered in this session.	Managing Large Datasets with MONGODB (OnlineVarsity) Session 5 & 6
9	MONGODB –TL9	All the topics as listed below from Session 7 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 7 – MongoDB Indexing</u> • Introduction to Indexes • Multi-key Index	Managing Large Datasets with MONGODB SG-Session 7 XP-Session 7 TG-Session 7

Session No.	Session Title	Session Details	Deliverables' Mapping
		- Hashed Index - Create Index - Drop Index	
10	MONGODB – TL10	All the topics as listed below from Session 8 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 8 – Replication and Sharding</u> - Replication - What is replication? - Deployment Architecture - Replication Commands and Methods - Implement Replication Using an Example - Sharding - What is Sharding? - Cluster Components - Shard Keys - Sharding Commands and Methods - Implement Sharding Using an Example	Managing Large Datasets with MONGODB SG-Session 8 XP-Session 8 TG-Session 8
11	MONGODB – TL11	All the topics as listed below from Session 9 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 9 – Transaction Management</u> - Introduction to MongoDB Transactions - Transaction Properties - Transactions and Sessions - Working With Transactions	Managing Large Datasets with MONGODB SG-Session 9 XP-Session 9 TG-Session 9
12	MONGODB – TL12	The Try It Yourself questions of Session 7 to Session 9 of <i>Managing Large Datasets with MongoDB</i> book should be covered in this session.	Managing Large Datasets with MONGODB Session 7-9
	S3	All the Assignments from Session 7 to Session 9 of <i>Managing Large Datasets with MongoDB</i> course should be covered in this session.	Managing Large Datasets with MONGODB (OnlineVarsity) Session 7-9

Session No.	Session Title	Session Details	Deliverables' Mapping
13	MONGODB – TL13	All the topics as listed below from Session 10 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 10 – MongoDB Tools</u> • Connect MongoDB Compass with MongoDB deployment • Manage Database Using Compass • Install MongoDB BI Connector • Connect MongoDB BI Connector with MongoDB deployment • Create a system Data Source Name • Manage Data Between MongoDB collection and Microsoft Excel	Managing Large Datasets with MONGODB SG-Session 10 XP-Session 10 TG-Session 10
14	MONGODB – TL14	All the topics as listed below from Session 11 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 11 – MongoDB Cloud</u> • Get Started with Atlas • Access MongoDB Atlas Cluster • Import Data into Atlas Cluster From MongoDB Instance • Export Data from Atlas Cluster Into MongoDB Instance • Administer MongoDB Cluster	Managing Large Datasets with MONGODB SG-Session 11 XP-Session 11 TG-Session 11
15	MONGODB – TL15	All the topics as listed below from Session 12 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 12 – MongoDB Database Connectivity with Python</u> • Overview of connecting Python with MongoDB • Install PyMongo driver • Create a Database and Collection Using Python • Insert Documents Using Python • Query Documents Using Python	Managing Large Datasets with MONGODB SG-Session 12 XP-Session 12 TG-Session 12

Session No.	Session Title	Session Details	Deliverables' Mapping
16	MONGODB – TL16	The Try It Yourself questions of Session 10 to Session 12 of <i>Managing Large Datasets with MongoDB</i> book should be covered in this session.	Managing Large Datasets with MONGODB Session 10-12
	S4 & S5	All the Assignments from Session 10 to Session 12 of <i>Managing Large Datasets with MongoDB</i> course should be covered in this session.	Managing Large Datasets with MONGODB (OnlineVarsity) Session 10-12

6. Library References

➤	MongoDB: The Definitive Guide by Kristina Chodorow and Michael Dirolf
➤	Mastering MongoDB 4.x by Wouter Thielen
➤	Data Modeling for MongoDB: Building Well-Designed and Supportable MongoDB Databases by Steve Hoberman, William Steve Hoberman

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